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Expertise in Tool Use Promotes Tool Embodiment

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Abstract

Body representations are known to be dynamically modulated or extended through tool use. Here, we review findings that demonstrate the importance of a user’s tool experience or expertise for successful tool embodiment. Examining expert tool users, such as individuals who use tools in professional sports, people who use chopsticks at every meal, or spinal injury patients who use a wheelchair daily, offers new insights into the role of expertise in tool embodiment: Not only does tool embodiment differ between novices and experts, but experts may experience enhanced changes to their body representation when interacting with their own, personal tool. The findings reviewed herein reveal the importance of assessing tool skill in future studies of tool embodiment.

Keywords: Tool use; Body representation; Embodiment; Training; Expertise; Tool incorporation; Perception

1. Introduction

Expertise in the use of a tool promotes an understanding of its affordances, or how a tool can augment one’s capacity for action (Gibson, 1979). Expertise is attained through extensive practice that yields an increased understanding of the effects one’s movement will have on the tool and subsequently how wielding the tool will alter one’s interactions with the

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environment. When a tool is used extensively, and its properties are so fully understood that the user thinks of the tool as an extension of their own body, the use of the tool is said to be “transparent” (Clark, 2003). The integration of a tool into one’s body representation has been demonstrated both behaviorally and physiologically (Berti & Frassinetti, 2000; Canzoneri et al., 2013; Cardinali et al., 2009; Maravita & Iriki, 2004; Maravita, Spence, Kennett, & Driver, 2002; Miller et al., 2018, among others) as one’s perceived body morphs to incorporate the tool. Gibson believed that the temporary attachment of tools to one’s body extends one’s capacity for action and blurs the boundary between body and environment (Gibson, 1979). In this narrative review, we discuss: (1) how tool embodiment is promoted by experience, and (2) how users’ subsequent perceptions are influenced through their experience with tool-enabled affordances.

Within an embodied perception framework, spatial perceptions are scaled by those aspects of the body that are relevant for intended actions (Proffitt, 2006). For example, a pebble selected for throwing would be scaled to the size of one’s hand, whereas an open field would be scaled to the amount of walking required to traverse it. As such, the environment is perceived in terms of one’s ability to act on it, with the available actions determined by the size and shape of one’s body and whether it has been augmented by tools. If the environment is perceived in terms of one’s ability to act, then objects that are within reach will appear closer than objects beyond reachable space. The use of a tool to expand the area within reach should thereby alter the distance perceptions of tool users, since the tool has expanded its wielder’s capacity for action. Participants in a series of studies by Witt et al. perceived a target that was within reach as closer when they held or imagined holding a tool, than the same target when it was beyond reach without the tool (Witt & Proffitt, 2008; Witt, Proffitt, & Epstein, 2005).

The findings of Witt and colleagues are representative of a sizable literature showing how wielding handheld tools influences behavioral, perceptual, and physiological responses to one’s immediately actionable surroundings, dubbed peripersonal space (Cardinali, Brozzoli, & Farnè, 2009; Rizzolatti, Fadiga, Fogassi, & Gallese, 1997). Recent conceptualizations of peripersonal space posit that it is a collection of fields proximal to the body where multisensory information is integrated, allowing the brain to detect and anticipate potential interactions between the body and external objects or events (Bufacchi & Iannetti, 2018; Hunley & Lourenco, 2018; Serino, 2019). The concept of peripersonal space, and its mutability in the case of tool use, emerged from a neurophysiological study of macaque monkeys. Two types of bimodal neurons in the intraparietal cortex were discovered: “distal type” and “proximal type” neurons. Distal type neurons respond to both somatosensory stimuli on the hand and to visual stimuli near the hand. In contrast, proximal type neurons respond to somatosensory stimuli on the shoulder and neck as well as to visual stimuli presented in the space that the monkey can reach without bending at the torso (Maravita & Iriki, 2004). Iriki and colleagues found that after training a monkey to use a tool to obtain food, the area where distal type neurons responded to visual and somatosensory stimuli expanded from the hand to include the tool. In addition, the proximal type neurons also expanded to respond to visual stimuli anywhere within the monkey’s new tool-expanded reach (Iriki, Tanaka & Iwamura, 1996). This suggests that these distal and proximal bimodal neurons are responsible for

combining information from proprioception and vision into a cohesive representation of the body's position in space and the space over which the animal can act (Graziano & Botvinick, 2002).

In lieu of directly recording from the intraparietal cortex of human participants, the cross-modal congruency task is a commonly used research paradigm in the investigation of multi-modal representations of peripersonal space (e.g., Maravita et al., 2002). In this task, participants sit with both hands outstretched in front of them. Their task is to respond as quickly as possible to vibrotactile stimulation that can occur on either their thumb or index finger of either hand. Their task is complicated by a visual distracter stimulus that can appear near the index finger or near the thumb of either hand. This combination of visual and tactile stimuli is believed to activate the distal type bimodal neurons documented in monkeys. When a visual distracter stimulus appears at a location incongruous with the vibrotactile stimulation, participants are much slower to respond to the vibrotactile stimulation. Participants are the slowest when the visual distracter appears near the same hand that receives the vibrotactile stimulation, but at different location on the hand (Maravita et al., 2002). It has been suggested that this delay is caused by the conflicting activation of bimodal neurons that respond to both visual and somatosensory information (Maravita & Iriki, 2004).

Given the finding that the distal type neurons of monkeys respond to visual stimuli surrounding a tool tip following tool training, the canonical delay in identifying the source of the vibrotactile stimulation at the hand should occur when a visual distracter stimulus is presented at the tool's tip. Indeed, participants who have practiced using a tool are slower to respond to a vibrotactile stimulus delivered to their hand when a visual distracter appears at the tip of their tool (Maravita et al., 2002). This suggests that, as in monkeys, active tool use expands human peripersonal space to include the tool, resulting in a delay identifying tactile information at the hand due to interference from processing the visual information at the tip of the tool. However, as with monkeys, this expansion of peripersonal space requires training with the tool: when participants passively held the tool, the crossmodal congruency effect vanished (Maravita et al., 2002).

2. Expert tool users

In the tool use studies discussed so far, the tool tasks were relatively simple, requiring touching a target (Witt et al., 2005) or crossing and uncrossing two held tools in a prescribed manner (Maravita et al., 2002). Nevertheless, participants in control conditions who did not practice with the tools did not demonstrate tool-use effects. To examine more complex tool use, either longer training sessions are required (10–15 minutes is typical), or experts who use their tool regularly in daily life, sport, or profession must be examined. While the tool use effects documented in novice participants who undergo training prior to the experiment do occur, they are also transient. In contrast, as we will show in this review, long-term tool users often have more pronounced tool use effects that can be elicited even in the absence of tool practice. These effects also persist and, in many instances, are dependent on whether or not the expert uses their own personal tool in the experiment.

One interesting category of individuals to examine is the blind, who habitually and functionally use a tool to expand their peripersonal space. Many blind people use a white cane, which allows them to integrate auditory and tactile information as they touch the area immediately in front of them with the cane's tip. Serino, Bassolino, Farne, and Ladavas (2007) demonstrated that blind cane users, experts in this tool's use, generate a new representation of multisensory peripersonal space in which the space around the tip of the cane has all the multisensory integrative properties found for the space surrounding the hand. In this study, participants were tasked with providing a verbal response to a weak vibrotactile cue that was interspersed with a strong vibrotactile cue presented as a distractor. Both cues always occurred on the right index finger and were always accompanied by an auditory stimulus. However, the source of the audio could be either near the hand or near the tip of the cane. Regardless of the source of the auditory stimulus, participants were told to ignore the distractor strong vibrotactile cue and to respond as quickly as possible to the target weak vibrotactile cue. Blind cane users responded faster to the target vibrotactile cue at the hand when the sound was presented near the tip of their cane than when it originated close to their hand. In other words, auditory information at the tip of the cane facilitated the detection of weak haptic stimulation at the hand. This crossmodal facilitation effect was found without any tool practice session prior to the experiment. This indicates that for blind expert cane users, the allocation of peripersonal space to the tip of the cane for the purpose of multisensory integration is not only permanent, but it may even reduce the effectiveness of multisensory integration at the hand relative to integration at the tip of the cane (Serino et al., 2007).

For sighted participants who were novice cane users, detection of the target vibrotactile stimuli was fastest when the source of the auditory stimuli was near the hand, rather than the tip of the cane. However, following 10 minutes of blindfolded cane practice, sighted participants' reaction times did not differ depending on the location of the auditory stimulus. This finding suggests, as in previous tool studies, that auditory peripersonal space expands to include a cane following practice. However, the authors found that this expansion recedes to pretraining levels after a day without cane use (Serino et al., 2007). Notice that in the absence of daily cane use to navigate their environment, the crossmodal facilitation effect observed in novice cane users was not nearly as pronounced as it was for blind cane users. As the authors point out, for a blind cane user wishing to avoid collisions with external objects, the space at the tip of the cane is much more important than the area around the hand. Thus, the altered peripersonal space for expert cane users is a functional adaptation that ensures their ability to traverse their environment without collision.

Another population of individuals who depend on daily tool use to maneuver in their environment is paraplegic wheelchair users, individuals with lesions below the seventh cervical spinal cord segment who experience sensory motor deficits in their lower but not upper limbs. To test how wheelchair embodiment affects distance perception, Scandola et al. recruited 20 wheelchair users and asked them to estimate the distance (in cm) to virtual flags presented at the top of ramps of varying lengths and inclinations. Participants viewed the flags through a virtual reality head-mounted display while seated in either their own personal wheelchair, or a generic wheelchair of the type used in airports and hospitals (Scandola et al., 2019). This distance estimation paradigm is based on work that indicates people scale distances by

how much effort would be required to walk a particular distance (Zadra, Weltman, & Proffitt, 2016). For wheelchair users, increasing both the distance to the flag and the inclination of the ramp should make the ascent to the flag more effortful and thus the distance estimates to such flags should increase. In fact, Scandola found exactly this pattern, but only when the wheelchair users were seated in *their own personal wheelchair*. Although participants seated in generic wheelchairs still judged distances to be greater when the flags were further away, there were no significant differences between distance estimates at different inclinations. This indicates that familiarity with and ownership of the tool is important for wheelchair users to perceive distances in an embodied manner. Wheelchair users report subjectively that their personal wheelchair is part of their bodily self (Papadimitriou, 2008), which may explain why the generic wheelchair interferes with distance estimations, even though wheelchair users have substantial practice using nonpersonal wheelchairs in situations like airports or hospitals.

Other expert tool users (such as musicians with bows and drumsticks or athletes who use rackets, clubs, or fencing swords) also report a subjective feeling of completion or wholeness when using the tools of their trade, and in particular, their own personal implement. When asked to discuss using a tennis racket that does not belong to him, professional tennis player Andre Agassi said it feels like “playing with a broomstick....as if I am playing left-handed, as if I’ve suffered a brain injury. Everything is slightly off” (Agassi, 2009; as cited in Biggio, Bisio, Avanzino, Ruggeri, & Bove, 2017). Using an adaptation of Serino et al.’s multimodal integration task described above, Biggio et al. determined that expert tennis players are faster at responding to vibrotactile stimulation at the hand when an auditory stimulus is delivered to the tip of *their own personal tennis racket* as compared to a racket belonging to another player (Biggio et al., 2017). Likewise, fencers holding their *personal epee* are also faster to detect a haptic stimulus on their hand when it occurs in synchrony with an auditory stimulus presented at the tip of their sword than when an auditory stimulus sounds at the tip of a generic fencing epee (Biggio, Bisio, Avanzino, Ruggeri, & Bove, 2020). This suggests that experienced athletes are sensitive to minute differences between the tools of their trade and are better able to incorporate into their peripersonal space the tool they have spent hours using.

Furthermore, a distinct pattern of motor-evoked potentials (MEPs) emerges when recorded from the hand and forearm of expert tennis players tasked with imagining executing a tennis forehand as opposed to a ping-pong forehand. The same procedure applied to novices does not result in a differentiable pattern of MEPs for tennis versus ping-pong forehands (Fourkas, Bonavolont, Avenanti, & Aglioti, 2008). Indeed, expert but not novice tennis players agree with the statement “the [tennis] racket is an extension of my hand,” and that they can readily invoke clear, vivid kinesthetic imagery of playing tennis (Fourkas et al., 2008; p. 2386). Similarly, researchers found that expert badminton players have a distinct pattern of MEPs recorded from their hand when imaging a badminton serve while holding a badminton racket. If instead the athletes held a weight- and length-matched plastic rod, the pattern of MEPs recorded during their motor imagery of a badminton serve did not differ significantly from the pattern recorded from novices imagining a badminton serve while holding either a racket or the control rod (Wang et al., 2014). These results indicate that the specificity of gesture and implement are of the utmost importance to expert tool users. Moreover, Biggio et al.

recently found that elite fencers also have a distinct pattern of MEPs recorded from the muscles responsible for controlling wrist movement when visualizing fencing-specific actions, but only when holding their own personal equipment, not when empty handed, nor when holding a generic epee (Biggio et al., 2020).

The fact that the muscles responsible for wrist movements respond differently to an imagined fencing scenario depending on whether an elite fencer is holding their personal epee as compared to a generic sword indicates that the exact shape and dimensions of a tool may be of greater importance than researchers had previously imagined. These findings may also hint at the notion that the personal tools of experts may be more strongly embodied than the simple tools used in classic studies of peripersonal space expansion. One way to shed light on this notion is to investigate tool use with research paradigms typically employed to assess body ownership.

3. Body ownership

The notion that tools are “*ready-to-hand*” or seamlessly incorporated into the wielder’s bodily dynamics appears in philosophy (Heidegger et al., 1962), but determining whether this is the case empirically requires creative experimental designs. One method for studying body ownership is to identify measurable differences in responses to a person’s own body as compared to other stimuli. The body view enhancement effect provides just such an opportunity. In the original study of this effect, participants were asked to lift their finger in response to a projected light that appeared either on their finger or 5, 10, or 15 cm to the right or left of their finger. Reaction times were up to 40 ms faster when the stimuli appeared on the finger than anywhere else (Hari & Jousmäki, 1996). Subsequent refinements of the body view enhancement paradigm reliably show a more rapid detection of visual targets presented on the hands, other body parts, or next to the hand (Dufour & Touzalin, 2008; Reed, Grubb, & Steele, 2006) than the same target presented in the absence of a body part. Moreover, visual stimuli presented directly on the hand are detected more rapidly than those presented on wooden hand-shaped objects (Whiteley, Kennett, Taylor-Clarke, & Haggard, 2004). Even more striking is the finding that this pattern persists when participants view video recordings of visual targets appearing on hands, but only when the participant believes that the hand seen in the video is in fact their own (Whiteley, Spence, & Haggard, 2008). This line of reasoning has been applied to studies of tool embodiment. In concert with the notion that tools extend peripersonal space to include the area surrounding the tool, the body enhancement effect also speeds the detection of visual targets presented on or near the functional portion of a hand-held tool, but only after the participant has undergone a training session with the tool (see Brown & Goodale, 2013, for review). Tellingly, wheelchair users respond equally fast to visual stimuli presented on their own, personal wheelchairs and their upper body. This response is faster than the response to stimuli presented identically on a generic wheelchair, or even on their upper body *when seated in a generic wheelchair* (Scandola et al., 2019).

Another common method for studying body ownership is to use illusions that alter a participant’s perception of their own body. Embodiment effects characterized by rubber hand

illusion (RHI) (Botvinick & Cohen, 1998) studies are commonly understood to depend on the subjective feeling of body-ownership rather than changes to one's peripersonal space. In the classic RHI, simultaneous visuo-tactile stimulation of a rubber hand and the participant's hidden hand induces feelings of ownership of the rubber hand (Botvinick & Cohen, 1998; Tsakiris, 2016; Tsakiris & Haggard, 2005). A commonly used behavioral proxy to assess the strength of the illusion is the change in the felt position of the hand known as proprioceptive drift: stronger subjective ownership of the rubber hand coincides with the feeling that the participant's real hand is located closer to the rubber hand (Tsakiris, 2010; Tsakiris & Haggard, 2005). People who experience the RHI illusion do not report feeling as though they have three hands, but rather that the rubber hand has replaced their hand (Longo, Schüür, Kammers, Tsakiris, & Haggard, 2008). This indicates that the illusion is not expanding the body representation as is thought to happen during tool use, but rather directly incorporating the rubber hand into the body representation by supplanting the limb receiving the tactile stimulation.

Even though participants report feeling as if the rubber hand has become a part of their body, the reaching actions of participants who experience proprioceptive drift following the RHI remain accurate (Kammers, de Vignemont, Verhagen, & Dijkerman, 2009). In other words, even though they report feeling as though their hand is located closer to the rubber hand, they can still accurately reach and grasp an object with the hand that was supposedly replaced by the rubber hand during the illusion. This finding can be directly contrasted with tool use paradigms that demonstrate using tools will alter the kinematics of reach to grasp movements (Cardinali et al., 2011, 2012; Martel et al., 2019). This suggests that although tool embodiment expands the body representation and subsequently alters how one reaches and grasps objects, embodying a rubber hand is less associated with changes to perception and action than are most tool embodiment paradigms.

Adopting a novel approach, Weser et al. examined whether the RHI paradigm could be used to examine tool-use effects. In one study, participants held a pair of chopsticks, viewed a rubber hand holding chopsticks, and received tactile stimulation at the tips of their chopsticks. Viewing identical stimulation on the rubber hand's chopsticks caused participants to report that their hand felt like it was closer to the rubber hand. Interestingly, motor experience using the tool impacted the strength of the illusion, suggesting a role of the action-mediating body representation as well (Weser, Finotti, Costantini, & Proffitt, 2017). Cardinali and colleagues conducted a similar experiment in which a mechanical grabber and the participants' unseen hand received simultaneous visuotactile stimulation. In this "toolish hand illusion," participants reported that their hand felt closer to the tool, that they phenomenologically experienced the tool as being their hand, felt that the touches on their hand originated on the tool, and finally showed an increase in arousal as measured by Galvanic skin response to a threat directed toward the tool. These three indicators of tool embodiment emerged even before participants actively practiced using the grabber tool and were not enhanced by prior tool practice (Cardinali et al., 2021). In the context of this review, the lack of a practice effect is surprising. There were, however, numerous methodological differences between Cardinali et al.'s study and that of Weser et al., perhaps most importantly was that the simultaneous stimulation was applied to the visually apparent tool and the participant's unseen hand rather

that an unseen handheld tool. The illusory effect was of the tool supplanting the hand, not augmenting its action capabilities. Future work examining the “toolish hand illusion” can elucidate the importance of motor experience in studies of illusion-based tool embodiment.

In contrast, practice using a tool to achieve a productive end is frequently cited in the tool use literature as a necessary component for a modification of peripersonal space and tool embodiment to occur (e.g., Brown, Doole, & Malfait, 2011; Maravata & Iriki, 2004; Witt et al., 2005). For example, Brown et al. demonstrated that the adaptation of the motor system, as opposed to visual familiarity with the tool, is necessary for changes in peripersonal space to take place. Active training participants completed a task with a reach-extending tool that had novel mass distribution, while passive training participants used the same tool with the weight of the tool and the participant’s arm entirely supported in a mechanical device. The active training participants were much quicker at detecting targets that appeared near the tip of the tool than were passive training participants. This suggests that when divorced from motor control, the visual experience of tool-wielding is not enough to induce tool-related spatial adaptation (Brown et al., 2011). Costantini et al. demonstrated that observing tool actions extends the representation of reachable space, but only when the observer is also holding the tool (Costantini, Ambrosini, Sinigaglia, & Gallese, 2011), highlighting again the importance of tactile experience. Similarly, in a mirror illusion paradigm, participants holding identical tools in each hand experience a kinesthetic illusion such that their stationary left arm feels as though it is moving in synchrony with the reflection of the tool held in their right hand. This illusory movement is experienced even when the moving right hand is occluded from view, suggesting that the visual cues of the moving tool alone are sufficient to induce the classic kinesthetic mirror illusion (Guerraz, Breen, Pollidoro, Luyat, & Kavounoudias,). Martel and colleagues (2019) confirmed that reach to grasp movements are altered following blindfolded tool use, indicating that tool-specific motor activity alone is sufficient to cause changes to kinematic changes previously documented in sighted tool use paradigms (e.g., Cardinali et al., 2011, 2012).

However, experts in the aforementioned tennis and fencing studies did not take part in the typical training regime used to elicit the body enhancement effect in other tool studies. By the very nature of their expert athlete status, they have *already* received extensive tool-training and so the enhanced detection of stimuli presented near their personal tools occurs just as naturally as it does on their own hands. In Weser et al. (2017), skilled chopstick users experienced a stronger RHI than did unskilled users. Chopstick skill was measured by having participants use chopsticks to sort by color as many small plastic beads as possible into separate containers. The number of beads transferred in a 5-minute period was used as a proxy for chopstick skill. Weser et al. found that the more beads a participant transferred, the stronger their RHI experience. Similarly, Weser and Proffitt found only individuals who self-reported frequent tweezer use experienced a tweezers-RHI in which a pair of tweezers held in the participant’s unseen hand was stroked at the same time as a pair of tweezers held by a visible rubber hand (Weser & Proffitt, 2019). Greater chopstick skill or more tweezer use is likely accompanied by increased sensitivity to tactile stimulation at the tips of these tools. Indeed, recent convergent evidence from behavioral psychophysics, structural mechanics, and neuronal modeling demonstrates that tools are treated by the human nervous system as sensory extensions of the

body (Miller et al., 2018). Greater sensitivity likely leads to increased “transparency” of the tool (Clark, 2003), enhancing one’s ability to perceive the environment through the embodied tool.

4. Embodiment and measuring changes to the body representation

Another technique for assessing the embodiment of tools is the tactile distance judgment task (De Vignemont, Ehrsson, & Haggard, 2005; Taylor-Clarke, Jacobsen, & Haggard, 2004), in which participants must judge whether the distance between two points applied to one part of the body is the same as the distance between two points applied to another body part (the forearm vs. the forehead, for example). Since perceiving the size of an object touching the skin depends on an implicit representation of the body (Longo, Azañón, & Haggard, 2010), changes to the body’s action capabilities following tool use, such as an increase in the represented length of the arm following the use of a reach extending grabber, should cause a change in the perception of the distance between two points applied to the forearm, but not to the forehead, for example. Indeed, this is exactly what has been found: tools that extend one’s reach (such as mechanical grabbers) influence the wielder’s representation of the length of their arm, but not the size of their hand (Cardinali et al., 2009; Miller, Longo, & Saygin, 2014). In contrast, tools that expand the grasp of the hand, such as a device that lengthens the wielder’s fingers but not the wielder’s reach, specifically alter the implicit representation of the size of the hand, but not the length of the arm (Miller et al., 2014). Interestingly, when the same long-handled mechanical grabber tool is used to achieve a different functional end, such as batting objects hanging from a string versus picking up and moving objects on the floor, the resulting changes to the implicit representation of the length of the arm are also distinct: Individuals using a grabber to reach and grasp a distant object make judgments indicative of a longer perceived arm length, whereas those that use the same grabber to swing at an object dangling in front of them make judgments indicative of a *shorter* perceived arm length (Romano, Uberti, Caggiano, Cocchini, & Maravita, 2019). These findings demonstrate that following extensive practice in laboratory settings, tools specifically alter the representation of the body part that they are functionally augmenting, and that this alteration depends upon the type of action implemented during training.

To further understand how the body is represented, Longo and Haggard developed a procedure to measure the implicit representation of the hand whereby participants place their hands underneath a covered surface and point to the locations of 10 landmarks (finger tips and first knuckles) on their unseen hand (Longo & Haggard, 2010). Interestingly, healthy adults consistently and significantly overestimate the width of their hands and underestimate the length of their fingers (Longo & Haggard, 2010; Longo, Mancini, & Haggard, 2015). Following tool-training, Cardinali, Brozzoli, Finos, Roy, and Farnè (2016) determined that sticks attached to the thumb and index finger caused the implicit representation of the thumb and index finger to increase. Similarly, training with conventional pliers caused an increase in the represented length of all of the wielder’s fingers (Cardinali et al., 2016), again demonstrating surprising specificity in the manner in which tool use alters body representation.

Interestingly, 15 minutes of practice with a baseball glove has no effect on the implicit hand representation of novice baseball players (Coelho, Schacher, Scammel, Doan, & Gonzalez, 2019). The authors speculate that this does not follow the typical tool training pattern because catching a baseball is a skilled behavior and the use of the glove may not have facilitated the task as readily as a grabber aids a participant in picking up and moving objects far out of reach. In contrast to the null findings for novices, players with 8 or more years of intensive baseball experience had a significantly *decreased* hand width and finger length of their gloved left hand relative to their right hand, suggesting that when ballplayers accustomed to glove use are not wearing their glove, they feel as if their hand is *smaller*. Indeed, one of the baseball player participants in this study stated that “when I reach out to pick up a ball without my glove, I feel like my hand is tiny and useless” (Coelho et al., 2019, p. 144). The authors speculate that the perception that one’s hand is smaller may also confer a functional advantage by ensuring accurate positioning of the center of the hand relative to the incoming ball. This is akin to results of grasping studies in which participants employ a large hand aperture when they are uncertain of the target to increase the margin of error and a smaller aperture when grasping is easy and certain (Jakobson & Goodale, 1991). This explanation is in alignment with the notion that perception, extended broadly to include somatosensation, informs people about their opportunities for action. A baseball player wearing his own trusty glove feels as though their hand is complete and can catch an incoming flyball.

5. Conclusion

As in baseball, all sports require consistency in the successful performance of sport-specific skills. Studies have shown that the apparent size of a target in a skill-based activity, such as batting, is scaled to the performance of the athlete. For example, softball players with better batting averages see the softball as bigger than players with lower batting averages (Witt & Proffitt, 2005). Likewise, field goal kickers who are kicking better perceive the distance between the uprights as larger (Witt & Dorsch, 2009). Although these examples do not assess the effect of tools on performance, they allude to a similar pattern of results. Expertise and skilled behavior alter aspects of perception that reflect the increased capacity for action afforded by the expert’s level of skill. In target-oriented tasks, targets are perceived as larger by more skilled actors (Lee, Lee, Carello, & Turvey, 2012; Witt, Linkenauger, Bakdash, & Proffitt, 2008). In tasks requiring the judgment of some aspect of the body itself, experts with extensive experience using a tool perceive their arm as longer, judge their hand to be smaller in the absence of their tool, or are faster to identify targets appearing near their personal tools.

Perception relates the environment to people’s everchanging purposes and the embodied means by which these purposes are achieved. Elsewhere, we have argued that people turn themselves into walkers, throwers, graspers, and so on according to their purpose, and in so doing, they perceive the world in relation to what they have become (Proffitt & Linkenauger, 2013). Experts, by the very nature of their expertise, are readily able to turn themselves into a tennis server, badminton player, fencer, baseball catcher, or in fact live their lives as a cane or wheelchair user. Therefore, when these experts are assessed using the experimental

techniques developed to examine the influence of tools on nonexperts, the results are typically more pronounced and enduring, as described above. Experts often report that their specialized tools feel like a part of their body. Given that people perceive affordances—the possibilities for acting in one's surroundings—the world is scaled to that aspect of our body that is relevant for the execution of the desired action. In other words, one's purpose determines the bodily dimension—including augmentations provided by embodied tools—that serves as the perceptual ruler by which the environment is scaled. Expert effects in the tool-use literature signal the ways in which experts embody their tools in a manner more pronounced than has typically been documented in tool-use studies that do not employ experts.

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